



CANNABIS
SCIENCE
CONFERENCE
AN MJLIFE SOLUTIONS® EVENT

May 18-20, 2022

Long Beach Convention Center
Long Beach, California



Keith Bellingham

Scientist, Stevens Water Monitoring Systems

**Cannabis Fertigation Scheduling and
Control Based on In Situ Sensor Inputs**

www.CannabisScienceConference.com

Cannabis Fertigation Scheduling and Control based on In Situ Sensor Inputs

Investigators:

Keith Bellingham P. H.

Scientist/Hydrologist

Stevens Water Monitoring Systems, Inc

Wally Daniel

Operations Director

Scientific Officer

Folium Farms

Seth McInenly

Operations Manager

Folium Farms



www.stevenswater.com



Percent Water Content of Grow Media Electrical Conductivity of Grow Media

Research deployment in Clackamas Oregon

- The health of the roots of the cannabis plant is directly related to health of the crop.
- Monitoring the percent water content of grow media and electrical conductivity can give us a lot of insight into the health and yield of indoor grown cannabis.
- Based on the work, moisture sensors can have a high return on investment.



Goals of this research

- How can we interpret moisture and EC data in an easy way to make meaningful decisions about the crop?
- Are best management practices for the placement of the sensors?
- What would be the return on investment for monitoring EC and soil moisture.



Instrumentation and Devices



The Stevens HydraProbe Soil Sensor measuring:

- %Moisture Content
- Bulk EC
- Temperature



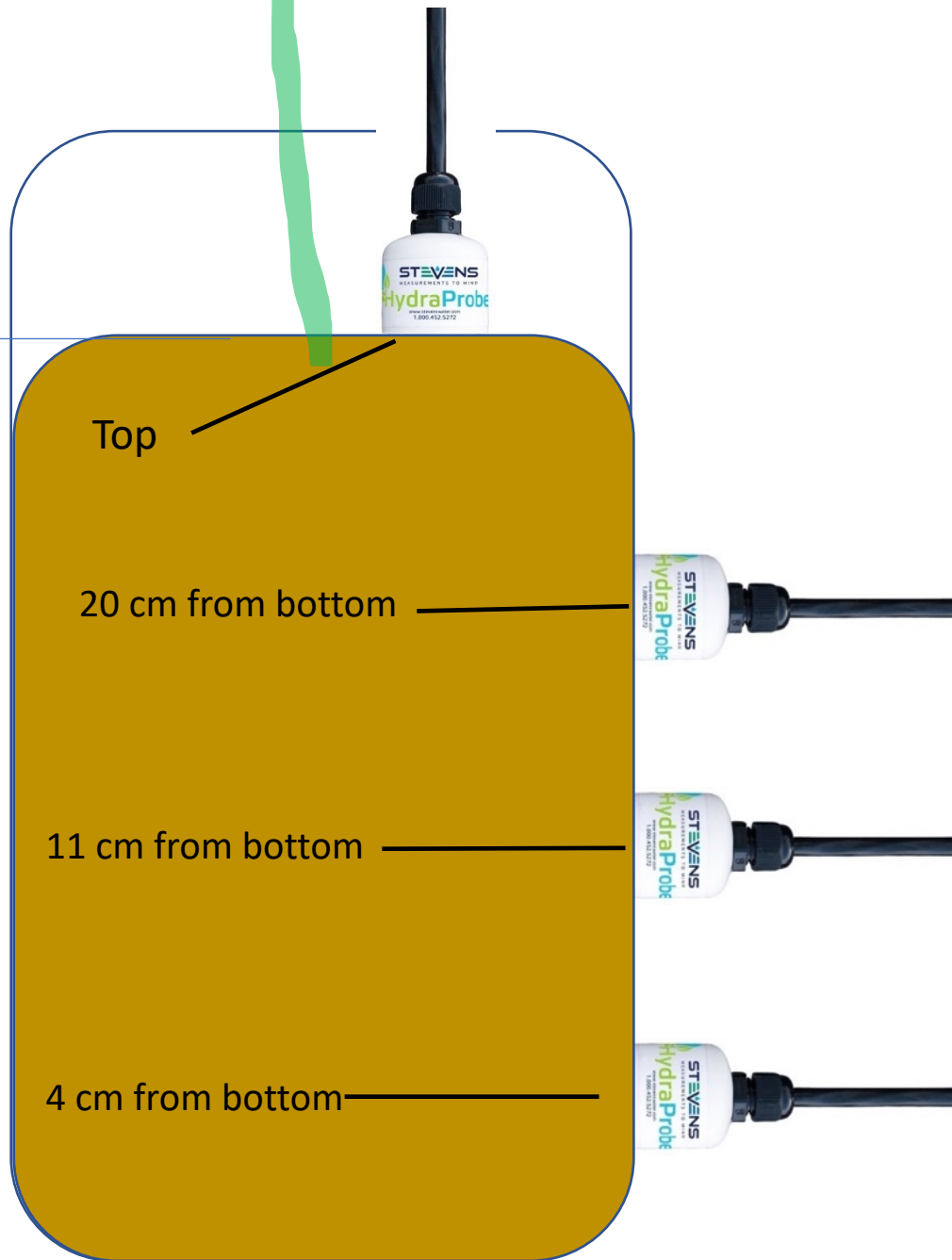
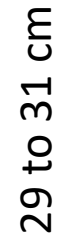
The Stevens Steelhead Logger



With Stevens-Connect as the SaaS.



35	35	35	35	35	35	35											
34	34	34	34	34	34	34											
33	33	33	33	33	33	33											
32	32	32	32	32	32	32											
31	31	31	31	31	31	31											
30	30	30	30	30	30	30											
29	29	29	29	29	29	29											
28	28	28	28	28	28	28											
27	27	27	27	27	27	27											
26	26	26	26	26	26	26											
25	25	25	25	25	25	25											
24	24	24	24	24	24	24											
23	23	23	23	23	23	23											
22	22	22	22	22	22	22											
21	21	21	21	21	21	21											
20	20	20	20	20	20	20											
19	19	19	19	19	19	19											
18	18	18	18	18	18	18											
17	17	17	17	17	17	17											
16	16	16	16	16	16	16											
15	15	15	15	15	15	15											
14	14	14	14	14	14	14											
13	13	13	13	13	13	13											
12	12	12	12	12	12	12											
11	11	11	11	11	11	11											
10	10	10	10	10	10	10											
9	9	9	9	9	9	9											
8	8	8	8	8	8	8											
7	7	7	7	7	7	7											
6	6	6	6	6	6	6											
5	5	5	5	5	5	5											
4	4	4	4	4	4	4											
3	3	3	3	3	3	3											
2	2	2	2	2	2	2											
1	1	1	1	1	1	1											
A	B	C	A	B	C	A	B	C									
R1			R2			R3			R4			R5			R6		



- 1) The Stevens HydraProbe Soil Sensor measuring:
 - %Moisture Content
 - Bulk EC
 - TemperatureLogging ever 15 mins.
- 2) 4 Different Depths
- 3) Five Replicates (20 sensors total)
- 4) Single Irrigation Zone Representing 630 plants
- 5) Sensors were installed in the “veg’ Stage and remained in place through the “flower” stage until harvest (60 days)

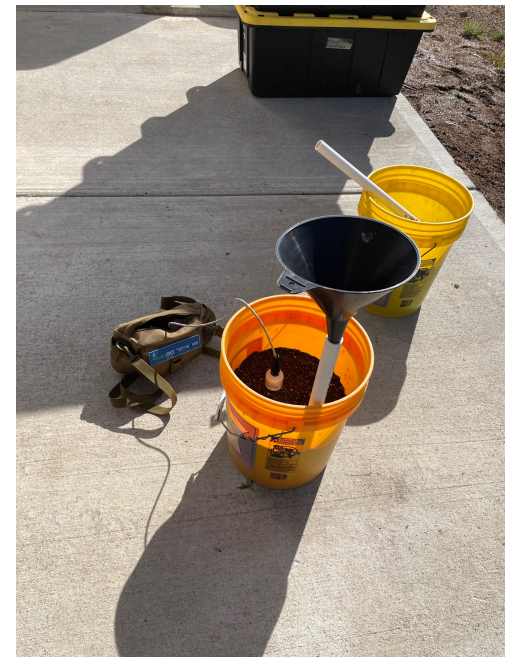
Irrigation Zone Description

- Grow bag size 29 x 12 cm – 12000 cm³
- About 630 plants per zone
- Grow Media is a coco coir, perlite (70:30)
- Delivery of fertigation water per zone in a batch Volume. Using a TGC PLC
- A batch volume is delivered 2 or 3 times per day over an 8 or 10 hour interval.
- 2 and 3 events per day are 250 and 450 gallons respectively.
- A 20% runoff is targeted.
- Two irrigation emitters per container



Physical Parameters

Parameter	Pre-Testing	In Situ Observations
Volume of Media in Bag	12,000 cubic cm	Same
Field Capacity of Media	40 to 42%	32 to 50%
Saturation of Media	60 to 62%	60 to 68%
Water Content of Stock media	20 to 24%	Same
Density of Stock Media @20%	0.40 to 0.42 g/cc	Same
Weight of Dry stock and bag	4890 to 4950 grams	Same





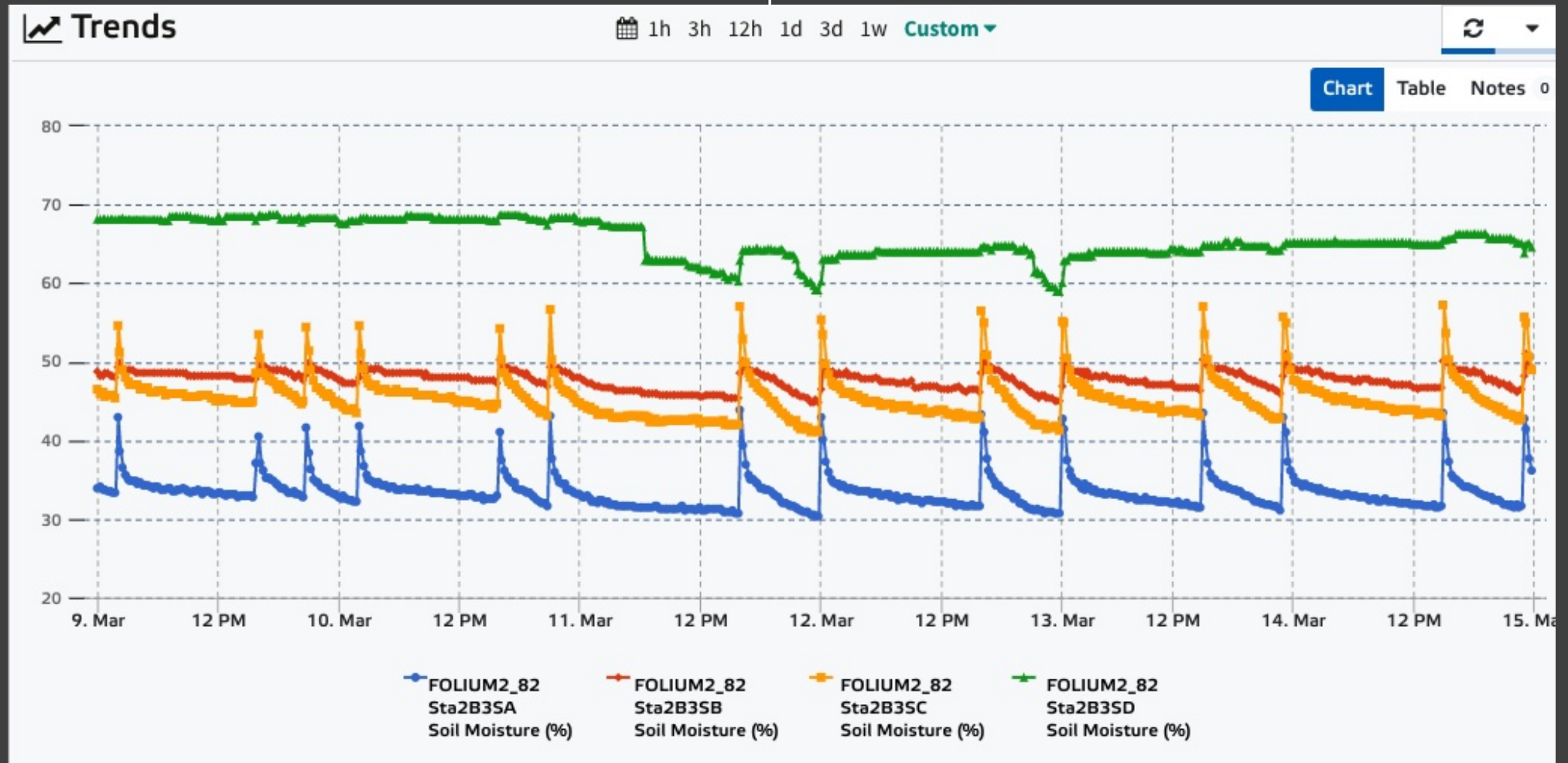
Images of setup

- HydraProbe sensors at 4 depths with 5 reps
- SCADA System by TGC.
- Reports to Stevens-Connect web service every 8 hours.

Trends in moisture as a function of depth.



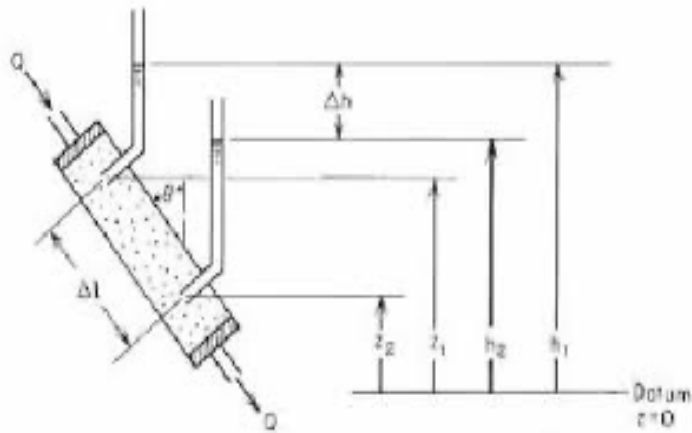
- The whole container wets up at once.
- The bottom stays near saturation at 65 to 68%.
- The moisture increase with depth



Saturated Flow Darcy's Law:

Saturated flow is almost all driven by gravity

$$v = -K_{sat} \frac{\Delta h}{\Delta l} \quad Q = -K_{sat} \frac{dh}{dl} A$$



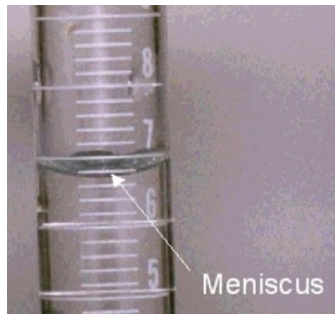
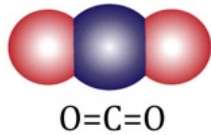
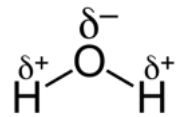
Unsaturated Flow: Richards Equation

Unsaturated Flow is driven by capillarity

$$\frac{\partial \theta(h)}{\partial t} = \frac{\partial}{\partial z} \left(K(h) \frac{\partial h}{\partial z} \right) + \frac{\partial K(h)}{\partial z} - S(h)$$

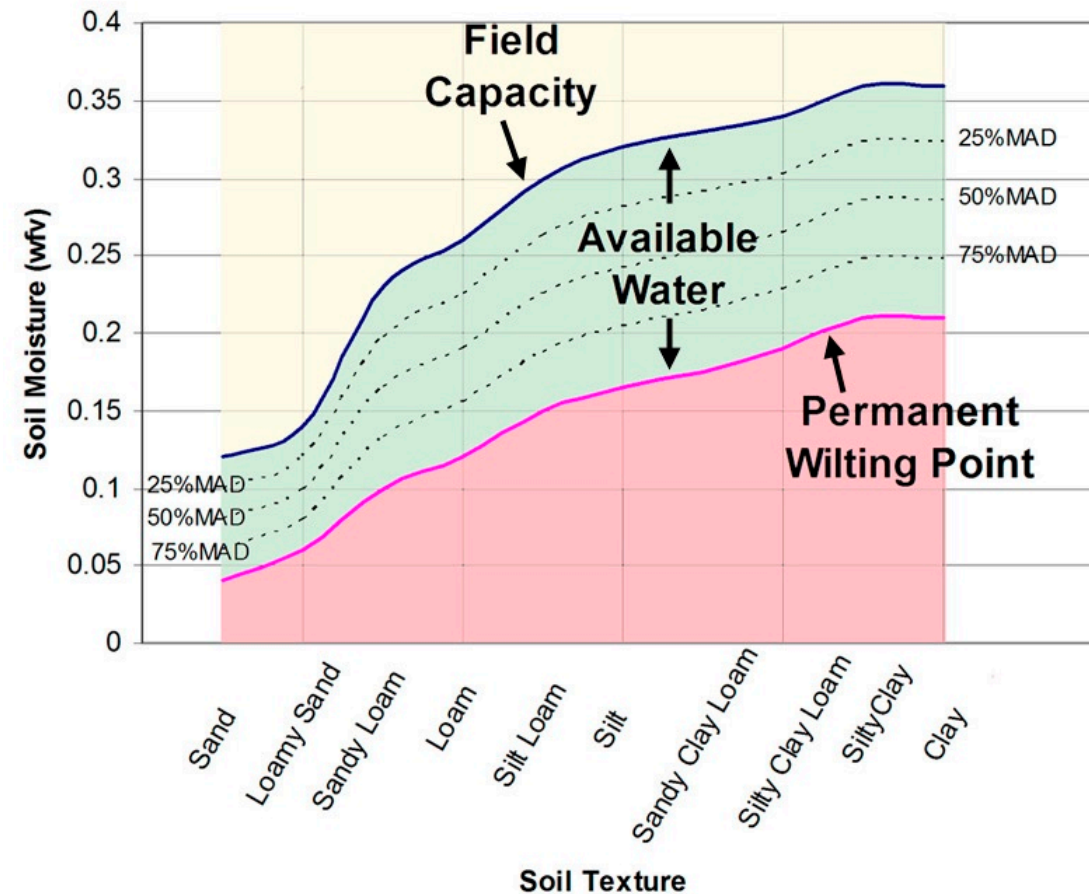
- $S(h)$ is ET,
- h is head, negative in unsaturated conditions
- z is depth
- θ is % moisture content
- K is the unsaturated hydrologic conductivity

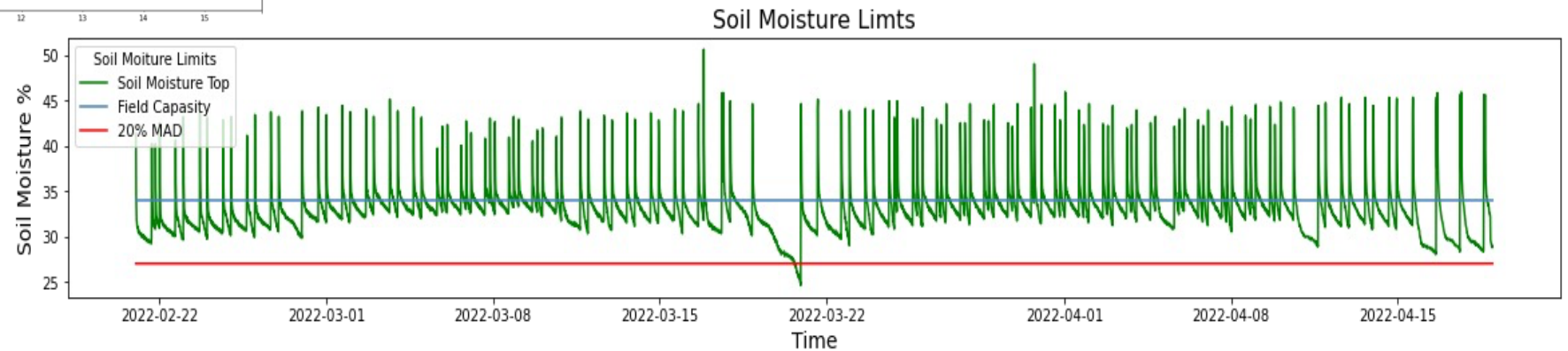
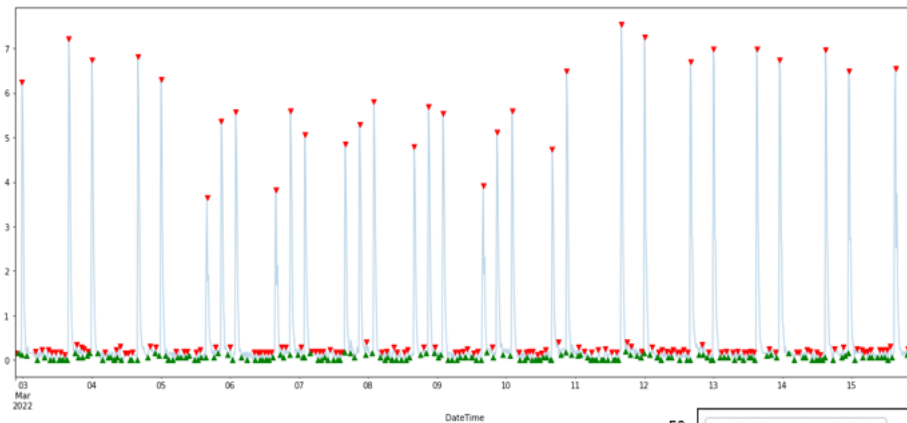
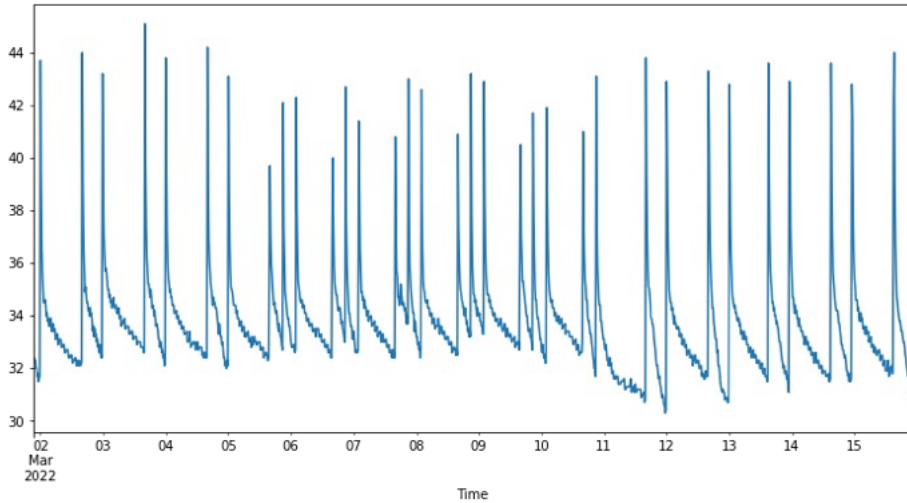
$$\frac{\partial \theta}{\partial t} = \text{Water Fluxes}$$



Field Capacity is an important hydrological threshold for the crop.

Soil Moisture Target





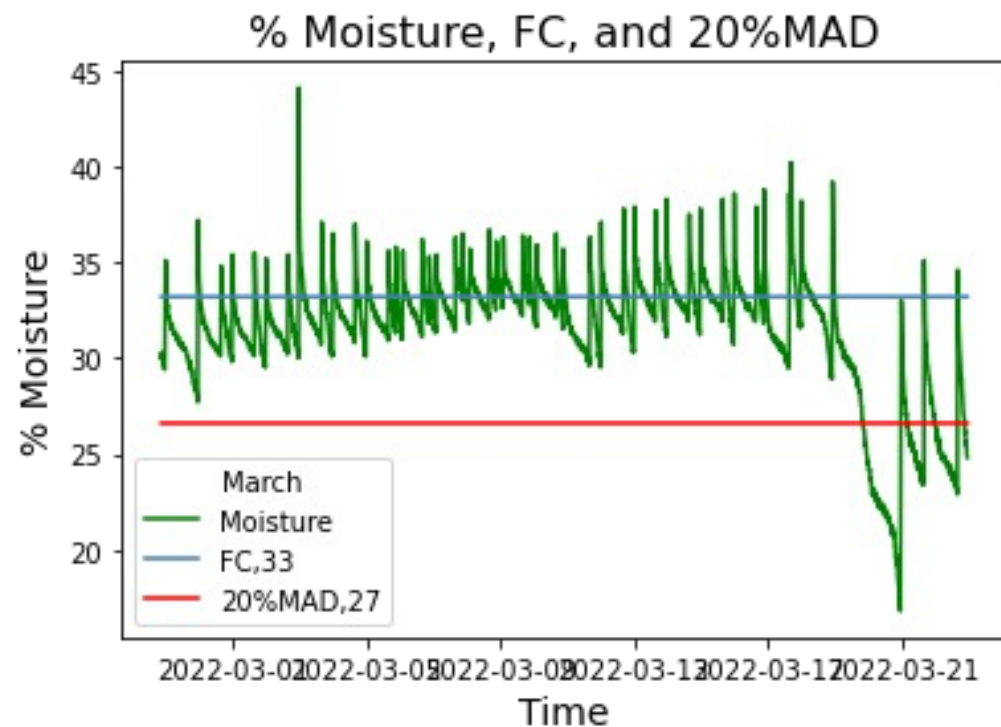
Data Science Tools

- Python, Pandas, and NumPy
- Forward rolling STD.
- Extreme point function in SciPy

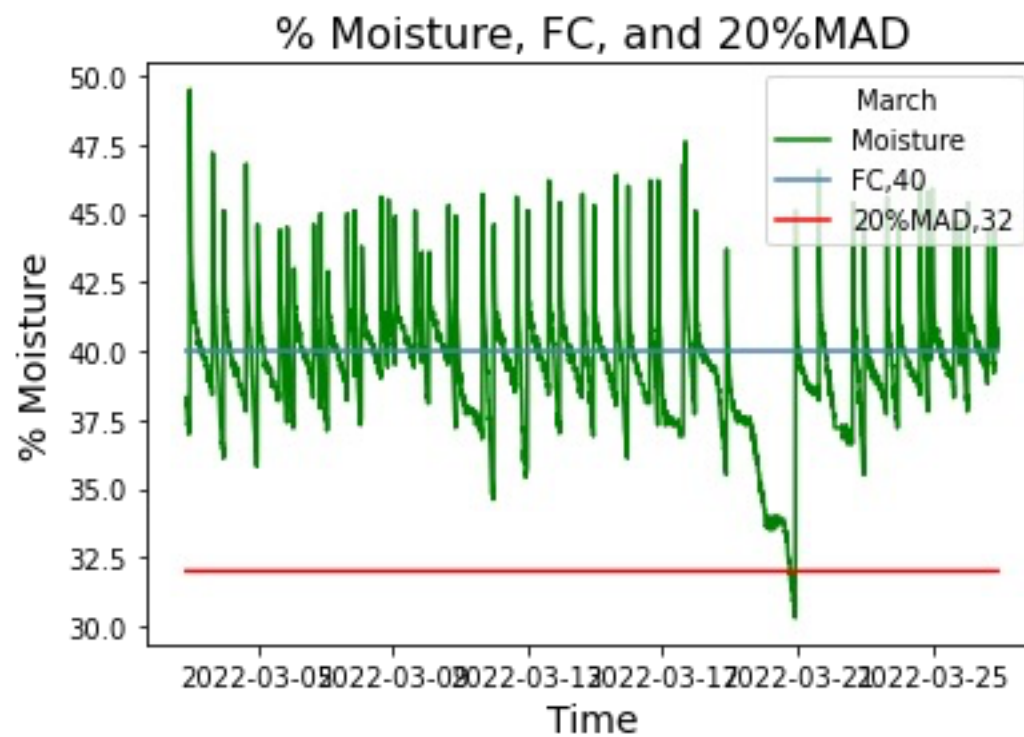
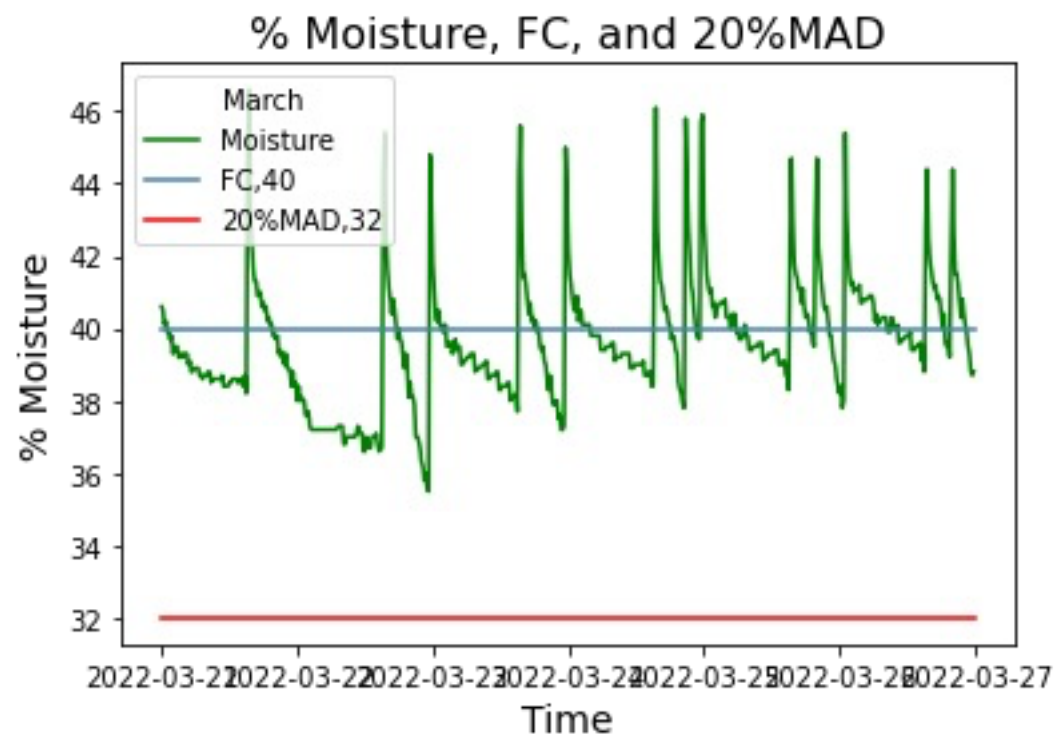
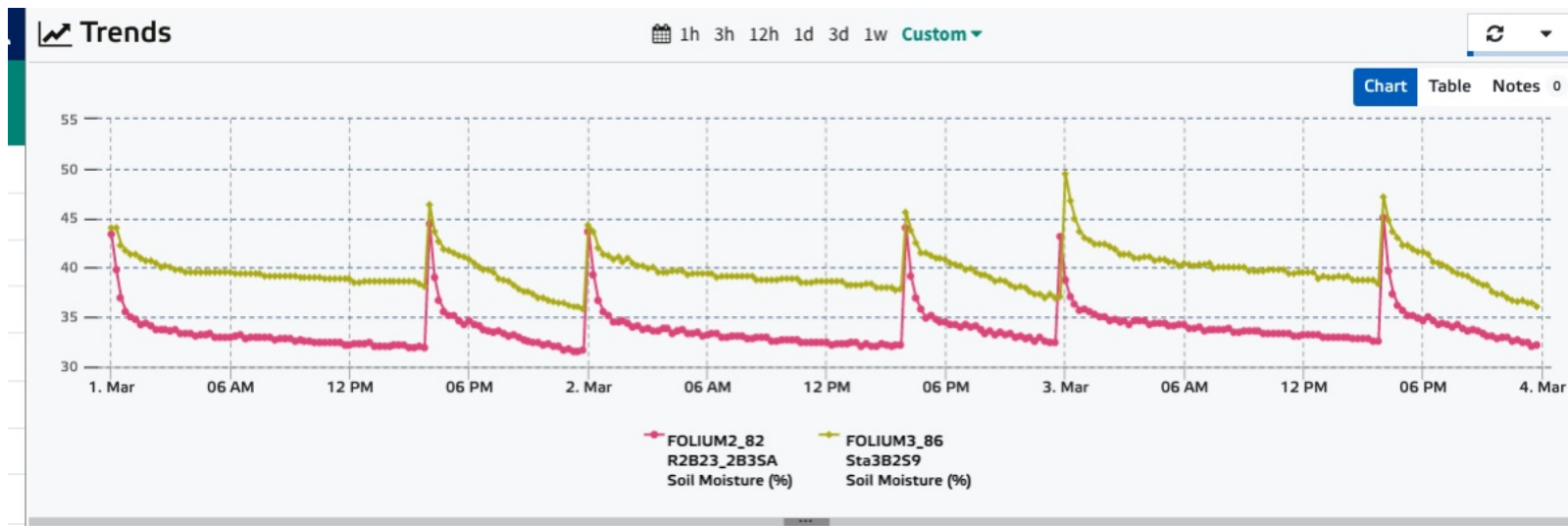


- After the Irrigation event 33.97%
- Std = 0.45%
- Before 32.85%
- Std = 0.92%
- 80% of FC or 20% MAD = 27%

Basically, an Algorithm can look at the moisture patterns measured with moisture sensor, determine the low moisture limit automatically. The cloud-based software can then send an alarm to the operator if the condition is reached.

[illegible]

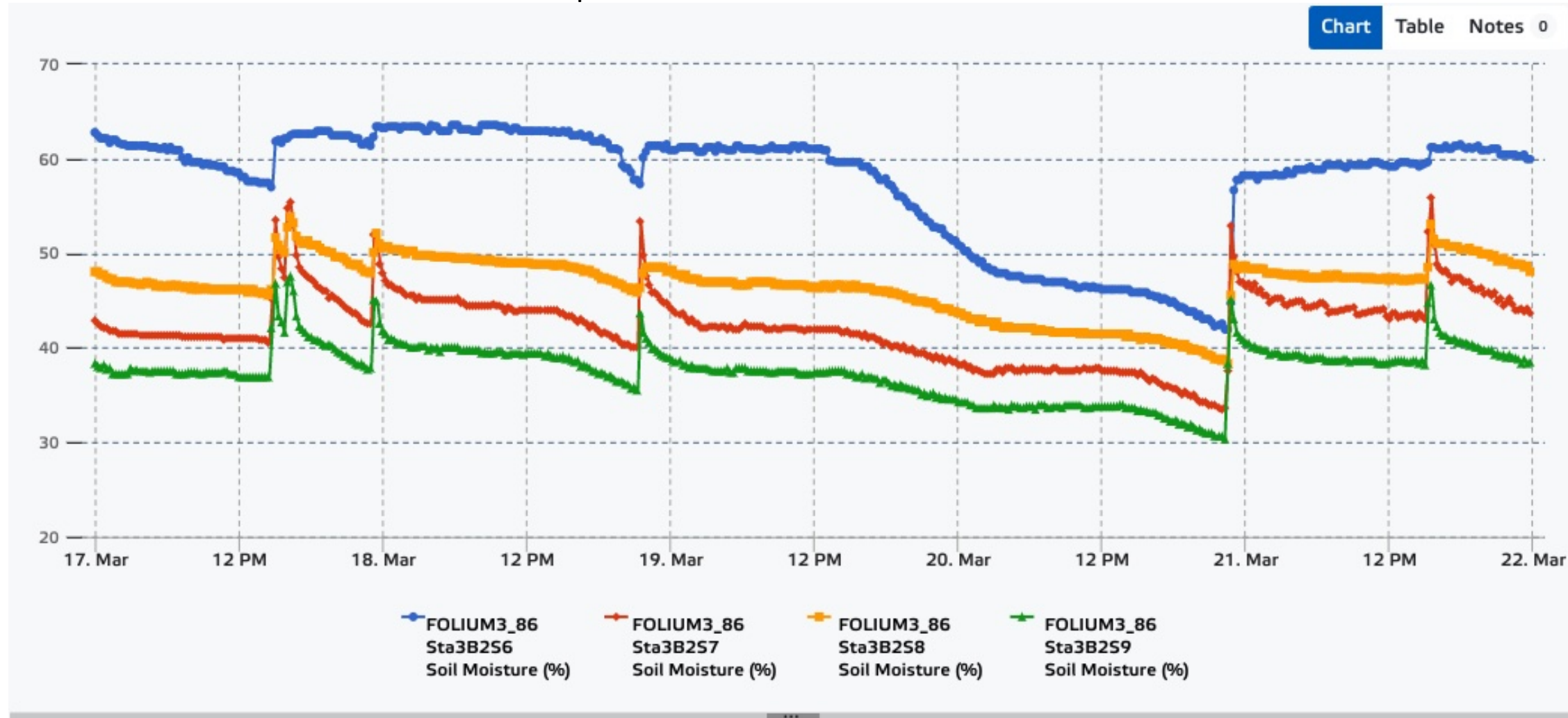
Alarms trigger based on a stress condition, [The dashboard, email and SMS Text Message].



A35

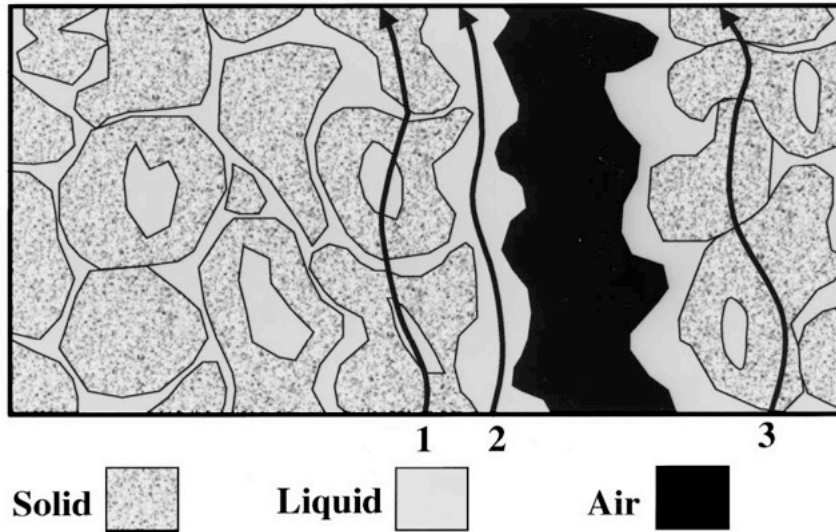
Crop Steering Dry-Back

- It is 15 hours before the moisture drops
- 48 hour dry back will drop the moisture level below is allowable depletion



Electrical Conductivity Pathways in Soil

Soil Cross Section



$$\sigma_p = \frac{\epsilon_{r,pw}\sigma_b}{\epsilon_r - \epsilon_{\sigma b=0}}$$

$$\epsilon_i = \epsilon_{rel} + \frac{\sigma_{dc}}{2\pi f \epsilon_v}$$

$$\sigma_{dc} = 2\pi f \epsilon_v \epsilon_i$$

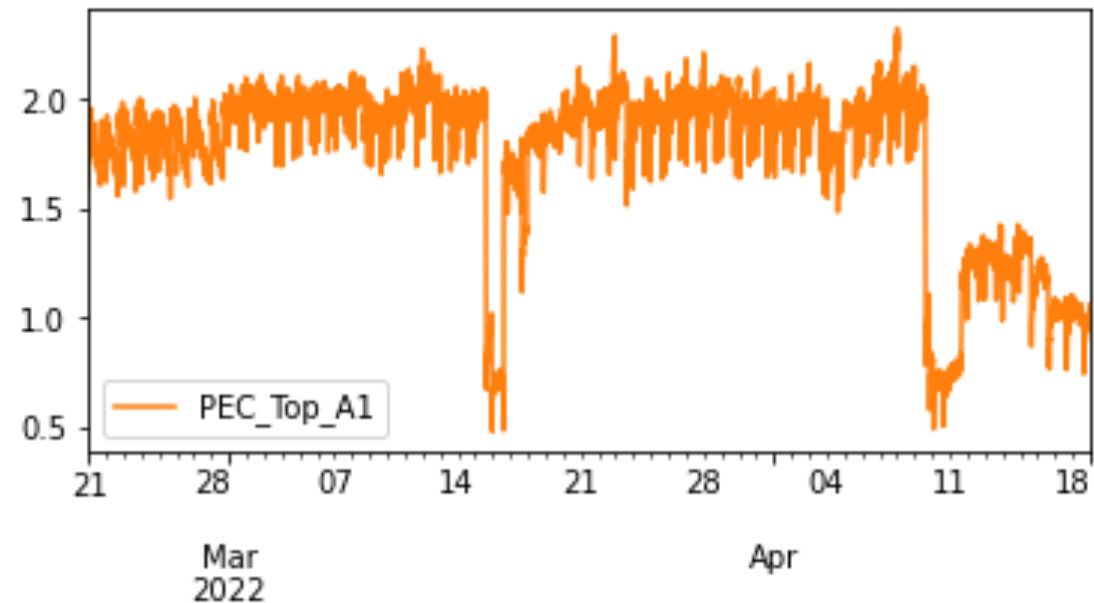
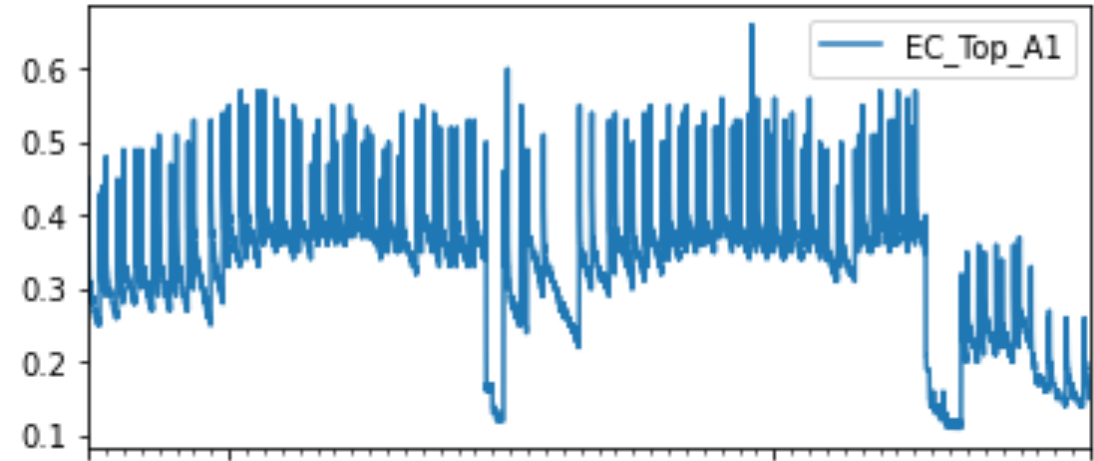
ϵ_i Imaginary Dielectric Permittivity

f Frequency (50 MHz)

ϵ_v Permittivity of a Vacuum

ϵ_{rel} Molecular Relaxation

σ_{dc} DC electrical conductivity



Return on Investments Model

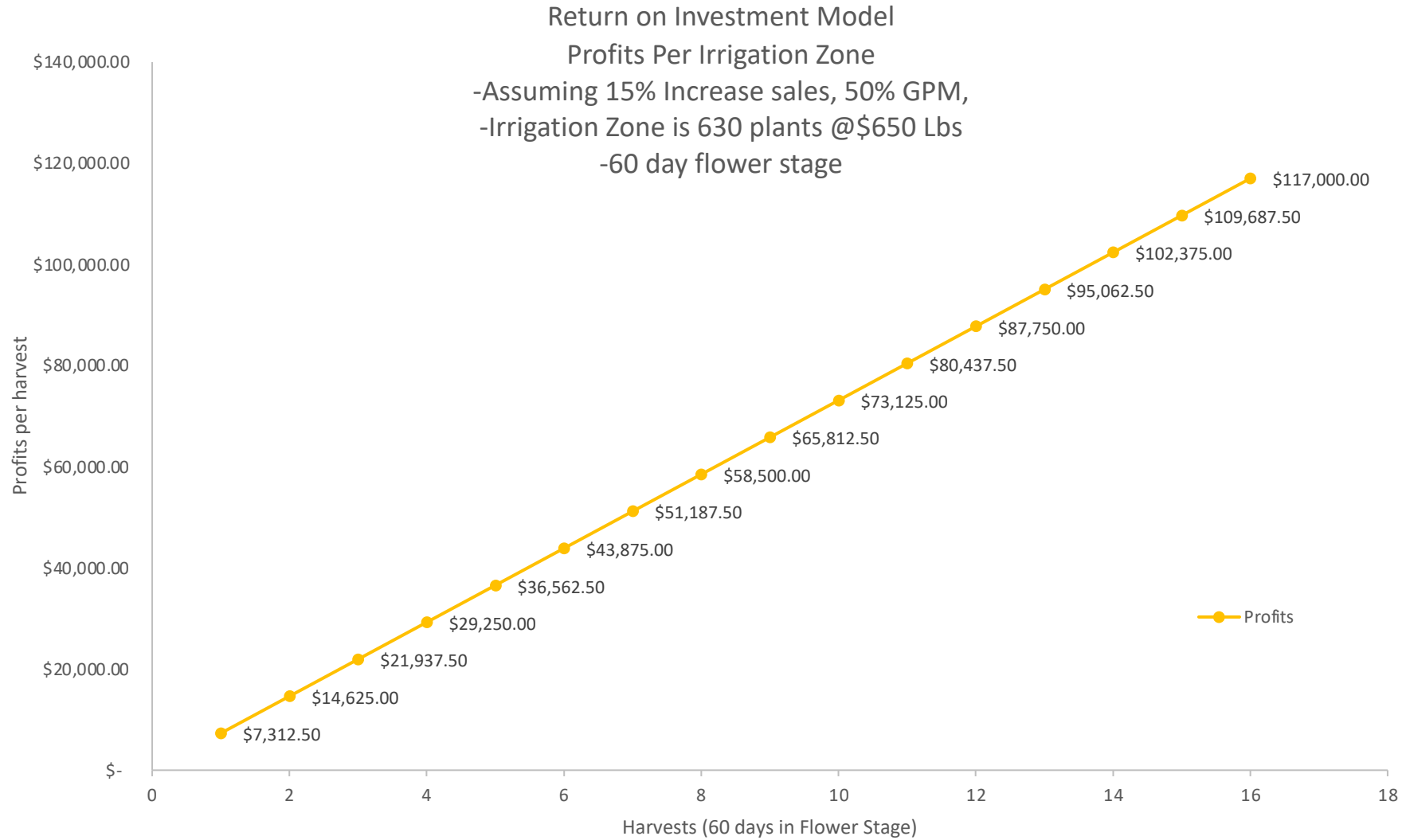
WITHOUT SENSORS for 20 Harvests for a zone of 630 plants			
Low	Mid	High	
90 lbs	150 lbs	210 lbs	lbs of sellable product per zone
25%	50%	25%	% of the time
5 harvests	10 harvests	5 harvests	20 harvests
\$292,500	\$975,000	\$682,500	Total Sales \$1,950,000* *@\$650 /lbs

WITH SENSORS for 20 Harvests for a zone of 630 plants			
Low	Mid	High	
90 lbs	150 lbs	210 lbs	lbs of sellable product per zone
0%	62.5%	37.5%	% of the time
0 harvests	12.5 harvests	7.5 harvests	20 harvests
	\$1,218,750	\$1,023,750	Total Sales \$2,242,500

$\$2,242,500 - \$1,950,000 = \$292,500$ for 20 Harvests

\$14,625 average additional sales per harvest

Monitoring moisture represents about 15% increase in sales



Conclusions

- Monitoring moisture with a lower limit value provides an immediate alarm of an irrigation failure.
- Monitoring the moisture around the roots can help make seasonal adjustments to the fertigation schedule.
- Because the ~80% of the root mass is in the top 6 inches, one sensor per container and one sensor for every 32 plants.
- In crop steering, a sufficient dry-back can be achieved from root zone monitoring.
- During a dry-back or during a period where the crop is stressed, based on the EC data, the crop increases its uptakes of nutrients out of the fertigation water.
- A moisture monitoring systems is predicted to have a 15% sales increase on total sellable product.



THANK YOU!

Keith Bellingham P. H.

kbellingham@stevenswater.com

503-445-8000

Stevens Water Monitoring Systems, Inc

www.stevenswater.com



Bibliography

- Šimůnek, J. M. (2008). Development and Applications of the HYDRUS and STANMOD Software Packages and Related Codes. *Vadose Zone J*, 587-600.
- Campbell, J. E. (1994). Dielectric properties and influence of conductivity in soils at one to fifty megahertz. *Soil Sci. Soc. Am. J.* 54:, 332-341.
- FAO-56. (1988). Crop and Evapotranspiration-Guidelines for Computing Crop Water Requirements Irrigation and Drainage. *Food and Agriculture Organization of the United Nations*.
- Hilhorst, M. A. (2000). A Pore Water Conductivity Sensor. *Soil Sci. Soc. Am.*, 64, 1922-1925.
- Richards, L. (1931). Capillary conduction of liquids in porous mediums. *Physics*, 1, 318-333.
- Seyfried, M. L., & Seyfried. (2005). Dielectric loss and calibration of the Hydra Probe soil water sensor. *Vadose Zone J.* 4:, 1070-1079.
- van Genuchten, M. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am.*, 44, 892-898.
- Warrick, A. W. (2003). *Soil Water Dynamics*. Oxford University Press.